MARIAM SALAH FAYED

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PROFESSIONAL SUMMARY

C/C++ Engineer with 4 years R&D at Siemens. Combines hands-on performance work on large simulation workloads with end-to-end ownership of customer-facing product areas.

EXPERIENCE

C++ Software Engineer – Siemens Digital Industries Software, Cairo, Egypt **Aug 2022 – Present**

Design Packager Tool (DP) | Unified design packaging solution serving all verification products

- Technical owner of Design Packager, a flexible common tool supporting multiple products and tool workflows; delivered customer-requested features end-to-end, from requirements and specifications through release in collaboration with Application Engineers, Product Engineers, and partner R&D teams.
- Integrated multiple tools with Design Packager and implemented ARM Architecture platform build support, extending its use across products and platforms.
- Mentored interns and junior engineers and wrote technical documentation used across teams.

QuestaSim – Siemens' HDL Simulator | Graph-based Elaboration (GBE)

- Responsible for the development in Cross Variant Connectivity area and delivered customer-requested features and performance enhancements end-to-end, from requirements and specifications through implementation and release while improving runtime efficiency, reducing memory consumption, and enhancing scalability across GBE workflows for complex designs.

QuestaSim Parallel Simulation Initiative

- Contributed to the design and implementation of multithreaded C++ prototypes for simulation, covering parallel data partitioning, synchronization, and result merging.
- Designed and implemented a GPU-based approach for a parallel simulation prototype, significantly improving performance compared with CPU-based and standard Questa execution while maintaining functional correctness.

Technical Stack: C++, C, Multithreading, CUDA, Jenkins

EARLIER EXPERIENCE

Application Development Intern, IBM **Mar 2022 – Jun 2022**
Supported and maintained production applications built on IBM integration tools.

Software Research Intern, Siemens **Sep 2021 – Dec 2021**
Evaluated TypeScript testing tools, selected Jest, and implemented unit tests for CI/CD pipelines.

Quality Assurance Intern, Siemens **Nov 2020 – Sep 2021**
Developed scripts and tools that automated testing activities and increased testing efficiency.

Artificial Intelligence Intern, Valoro **Aug 2020 – Sep 2020**
Worked across the SDLC, from requirements to deployment, on a full-stack AI virtual medical assistant.

TECHNICAL SKILLS

C++ & Software Design: Modern C++, C, object-oriented design, design patterns, refactoring, code review
Performance: POSIX multithreading, concurrency, synchronization, GPU programming, profiling, optimization
Tools & Platforms: Linux (x86-64, ARM), Git, Perforce, Make, Jenkins CI/CD, Coverity static analysis, Jira, Confluence
Scripting & Other Languages: Python, Tcl, Shell, SQL; familiar with Java and C#
EDA Domain: Questa simulation, SystemVerilog/VHDL mixed-language designs, profiling
AI & Automation: Familiarity with LLMs, AI agents, and generative AI tools

EDUCATION

B.Sc. in Computer Engineering, Faculty of Engineering, Ain Shams University **2017 – 2022**
GPA 3.8/4.0 (Excellent) ranked 3rd in class.

ADDITIONAL INFORMATION

Languages: Arabic (native), English (fluent)
Leadership: Student-activity leadership roles at ACES, MECA, and Egypt Career Summit (2018–2020)
Sports: Former professional basketball player, El-Shams Sporting Club (2013–2017)